

16-Jul — completed

All engine items are **live on the dev/test stack** (fc-estimate-dev.figitalabs.com) per the day's directive — flow work on the test surface, production untouched. Hospital_OS items are on the feature branch. Yesterday's 15-Jul round went to production in both repos last night.

Flow 2 — the flow, executed as an auditable SOP (the day's centrepiece)

- ☒ **POST /api/flow2/evaluate** — your flow note as a stateless step trail: payor → payor group + tariff → payor-aware family match → surgical/medical · daycare · robotic characterization (from THIS hospital's history only; a mandatory question wherever both sides exist, case counts on the option labels) → billing identification via the package master (inclusion/exclusion attached review-only) → your exact FC-historic fallback ladder with every rung visible → per-payor template summary
- ☒ **Pure-history numbers:** every bucket = P25/P50/P75 of the matched IP cases — no tariff/LOS/logic math; case-set filter chips (payor scope / daycare / robotic / care type) re-derive everything live; every number names its IP cases (drill-down with per-case payor/setting/robotic flags)
- ☒ **mode = logic / both** — the dissection layer: the real build runs beside the history (it receives only the audited decisions, never free text) and every bucket gets a verdict — "historically ₹36,871–₹49,208; logic produced ₹42,558 — within" — plus a gross verdict and a selectable logic room
- ☒ **Combo — a path per treatment:** multi-treatment wording gets per-treatment tabs, each with its own full trail, questions and numbers; billing shape identified ("2 packages" / "package + non-package"); combined P50 strip marked as an upper-bound reference
- ☒ **Two review surfaces:** the Hospital_OS admin "Flow 2" view (feature branch) AND a standalone page on the engine's own test frontend — **fc-estimate-dev.figitalabs.com/flow2.html** — verified by driving it in a real browser
- ☒ **374-case validation sweep** (every family × Cash + GIPSA + sampled Non-GIPSA, 604 evaluations, 215 questions answered): **zero data failures** — step order, payor resolution, question hygiene, quartile ordering, case-set filter obedience all held; gross quartiles independently recomputed from the returned IP lists matched exactly (the numbers provably ARE the history). 6 findings, all fixed same day. Report: [flow2-validation-16jul.md](#) (+ PDF)
- ☒ **Matching made deterministic-in-session and fast:** family match + package ranking cached per wording (answer round-trips 7s → 0.04s; mid-conversation family/package flips structurally impossible) with retry/backoff under transient AI flakes

Package-bill historic metrics (your morning ask #1)

- ☒ **fc.package_bill_bucket_metrics** — every billed line of every single-package bill classified into the estimate's own buckets (package line excluded; **implants split from pharmacy**; 176 package codes, 2,434 rows). What rides ABOVE the package, per bucket per payor group
- ☒ Rides the offer as **billed_actuals.bucket_extras**; the Historic-metrics panel gains a "Package bill" section (with-package total vs the actual final-bill band + the charged-above-package bucket table)
- ☒ GIPSA TKR proof: Procedure/OT extras P50 **₹1,21,300** (the robotic add-on), Implants ₹88k, PF/Pharmacy/Room small — the actual-vs-estimate gap explained by data
- ☒ Data flag for you: 973 billed admissions carry package names that aren't in the package master

Range / total clarity (your "clarity kahan pe hai")

- ☒ Audited every money surface; **the engine had zero inverted bands** — all display-side. "160 to 140" = the headline totals shown under an actuals band that never checked itself; "rows don't sum" = a range in the total cell + missing package/total rows
- ☒ Fixed: one shared bracket-rule implementation; every band now **labelled** (system range / past itemized bills / past package bills); preview total = exact sum with the expected range beneath it; bucket table proves base + buckets = headline to the rupee; and an explicit amber "past package bills ran ₹X–₹Y — this estimate is below that range" line (the conversion alert, finally visible). Audit: `range-audit-16jul.md`

Professional Fees fallback (your note ¶2)

- ☒ Room-matched PF: same room category + standard single-procedure case + bill within $\pm 15\%$ of the cohort P50 → that set's median PF, with the sample IPs and criteria shown. A recommendation rung — never a silent reprice. GIPSA TKR: 248 → 92 qualifying cases → PF P50 ₹56,584

Multi-treatment combos at intake (main flow)

- ☒ Resolve now detects combos and the Simple flow announces them in your words — "2-treatment combo: Lap Cholecystectomy (package SGA5166) + Inguinal Hernia Repair (non-package) — billing shape: package + non-package" — with a picker for which treatment this estimate builds. Detection + announcement; combined pricing stays the later phase (Flow 2 already paths each fully)

Small fixes from your testing

- ☒ Type-ahead search waits 1100ms (was firing per keystroke)
- ☒ Cash payor step no longer claims the insurer mapping — "Cash — the hospital's own TR1 (KIMS) tariff applies directly" (the decision was always correct)
- ☒ Downstream steps say "waiting on the answer above" instead of a false "no match" while a question is open

Deploys / state

- ☒ Engine dev/test stack: everything above live (flow2 API + page, bucket metrics, PF fallback, combo detect, caching)
- ☒ Hospital_OS: feature branch through `e5387dc` (Flow 2 view, combo UI, range fixes, debounce) — not merged to dev/main pending your Flow 2 review
- ☒ Engine production: untouched today per your "test version, not production" directive — promotes after your review

Waiting / next

- ☐ Your hands-on Flow 2 review → then we port the flow into the main estimate path
- ☐ Your call on the token ₹0/₹1 items (`token-rate-items-15jul.md`)
- ☐ NME design after your FC-staff sitting (manual entry live meanwhile)
- ☐ Whatever the hospital FC-module drill-down surfaces
- ☐ Combo interaction pricing (shared LOS/OT, package overlaps) — the agreed later phase
- ☐ Deviations for your blessing: robotic wording auto-decides robotic=yes; questions come one at a time; combo totals are a plain P50 sum